

SQL Practice — 20 New Questions

PostgreSQL — Company Database

BY Eng | Ahmed Salem

1) Employees Born Before 1965

Show the employees whose date of birth is before the year 1965.

Required (SELECT columns):

- fname
- lname
- bdate

Order the results from oldest to most recent by date of birth.

Concepts:

- WHERE
 - Date comparison
 - ORDER BY
-

2) Employees With Salary Between 25000 and 40000

Show the employees whose salary is between 25000 and 40000, inclusive.

Required (SELECT columns):

- fname
- lname
- salary

Use BETWEEN.

3) Employees From Specific Departments

Show the employees who work in departments: Department 4, Department 5.

Required (SELECT columns):

- fname
- lname
- dno

Use IN instead of writing multiple ORs.

4) Employees Whose Last Name Starts With S

Show the employees whose last name starts with the letter S.

Required (SELECT columns):

- fname
- lname
- salary

Use LIKE.

5) Employees With Salary Above 30000 and Born Before 1970

Show the employees whose salary is greater than 30000 and whose date of birth is before 1970-01-01.

Required (SELECT columns):

- fname
- lname
- bdate
- salary

Concepts:

- AND
 - Date comparison
 - Numeric comparison
-

6) Department Managers

Show the department name and the name of the employee who acts as manager of that department.

You'll need to join department and employee using mgrssn.

Required (SELECT columns):

- dname
 - fname
 - lname
 - mgrstartdate
-

7) Employees and Their Department Locations

Show employees along with the location of the department they work in.

You'll need to use employee, department, dept_locations.

Required (SELECT columns):

- fname
- lname
- dname
- dlocation

Note: if a department has more than one location, the employee may appear more than once.

8) Employees Working on ProductX

Show the data of employees who work on the project: ProductX.

You'll need to join employee → works_on → project.

Required (SELECT columns):

- fname
 - lname
 - pname
 - hours
-

9) Employees Working More Than 20 Hours on a Project

Show the employees who work more than 20 hours on any project.

Required (SELECT columns):

- fname
- lname
- pname
- hours

Order the results by hours from highest to lowest.

10) Total Hours for Each Employee

Calculate the total working hours for each employee.

Use SUM() and GROUP BY.

Required (SELECT columns):

- fname
- lname
- total_hours

Note: focus only on employees who exist in works_on.

11) Number of Projects for Each Employee

Show the number of projects each employee works on.

Use COUNT() and GROUP BY.

Required (SELECT columns):

- fname
 - lname
 - project_count
-

12) Employees Working on Exactly Two Projects

Show the employees who work on exactly two projects.

Think about using GROUP BY then HAVING.

Required (SELECT columns):

- fname
 - lname
 - project_count
-

13) Projects With Total Hours Greater Than 30

Show the projects whose total working hours is greater than 30.

Required (SELECT columns):

- pname
- total_hours

Concepts:

- SUM()
 - GROUP BY
 - HAVING
-

14) Departments With Average Salary Between 25000 and 40000

Show the departments whose average employee salary is between 25000 and 40000.

Required (SELECT columns):

- dno
- average_salary

Use AVG(), GROUP BY, HAVING, BETWEEN.

15) Department With the Highest Average Salary

Show the department with the highest average salary.

Required (SELECT columns):

- dname
- average_salary

Important: do not use window functions. Think about GROUP BY, AVG(), ORDER BY, LIMIT.

16) Employees Who Work on Projects Controlled by Another Department

Show the employees who work on a project that belongs to a department different from the department the employee works in.

Example idea: an employee works in Department 5 but works on a project that belongs to Department 4.

Required (SELECT columns):

- fname
- lname
- employee_department
- pname
- project_department

Concepts:

- Multiple JOINS
 - Comparison between columns
-

17) Employees Who Work More Than 10 Hours on Projects in Stafford

Show the employees who work more than 10 hours on projects located in Stafford.

Required (SELECT columns):

- fname
- lname
- pname
- hours

Concepts:

- Multiple JOINS
 - WHERE
 - Comparison
-

18) Employees Who Have Dependents

Show the names of employees who have dependents.

Each employee should appear only once, even if they have more than one dependent.

Required (SELECT columns):

- fname
- lname

Hint: think about DISTINCT and the tables employee, dependent.

19) Employees Who Have the Same Salary as Another Employee

Show the employees who have the same salary as another employee.

Example: if there are 3 employees with a salary of 25000, they should all appear.

Required (SELECT columns):

- fname
- lname
- salary

Important: use a SELF JOIN, and don't let an employee be compared to themselves.

Concepts:

- Self Join
 - Table Aliases
 - <>
 - =
-

20) FINAL CHALLENGE — Department Project Report

Build a report for each department that includes:

- Department Name
- Number of Employees
- Total Salary
- Average Salary
- Number of Projects
- Total Project Hours

Tables needed:

- department
- employee
- project
- works_on

Requirements:

- Join employees to departments
- Count the number of employees in each department
- Calculate the total employee salary in each department
- Calculate the average employee salary in each department
- Count the number of projects belonging to each department
- Calculate the total hours worked on each department's projects
- Group the results by department
- Order the departments by Total Salary from highest to lowest